The interaction between heterosis and hybrid incompatibilities

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Global settings, setting global font family (Default is Arial)

In[•]:=
```
SetDirectory[NotebookDirectory[]];
SetOptions[{Plot, ListPlot, ArrayPlot, ContourPlot, DiscretePlot3D, BarLegend,
   PointLegend}, LabelStyle → {Directive[FontFamily → "Latin Modern Math"]}];
```

Let the haplotype frequencies of ab, aB, Ab, and AB be described by the vector **x** = {x1, x2, x3, x4}. And linkage disequilibrium given by **D** = $x_1\, x_4 - x_2\, x_3$

In[•]:=
```
Clear["Global`*"]
x = {x1, x2, x3, x4} ;
DAB = x[[1]] * x[[4]] - x[[2]] * x[[3]];
(*m = m2 + m3;*)
(*Some conversion rules (Haplotype→Allele)*)
transferpRule =
  Solve[{pA == 1 - x1 - x2, pB == 1 - x1 - x3, ld == x1 * (1 - x1 - x2 - x3) - x2 * x3},
    {x1, x2, x3}] // Flatten;
alleletohaplo[{pA_, pB_, LD_}] :=
  {1 + LD - pA - pB + pA pB, -LD + pB - pA pB, -LD + pA - pA pB} // Simplify;
haplotoallele[x_] := {1 - x[[1]] - x[[2]], 1 - x[[1]] - x[[3]],
    x[[1]] (1 - x[[1]] - x[[2]] - x[[3]]) - x[[2]] × x[[3]]} // Simplify;
haplotogeno[x_] := {x[[1]] * x[[1]], 2 * x[[1]] * x[[2]], 2 * x[[1]] * x[[3]],
   2 * x[[1]] * (1 - x[[1]] - x[[2]] - x[[3]]) + 2 * x[[2]] * x[[3]], x[[2]] * x[[2]],
   2 * x[[2]] * (1 - x[[1]] - x[[2]] - x[[3]]), x[[3]] * x[[3]], 2 * x[[3]] * (1 - x[[1]] - x[[2]] - x[[3]]),
   (1 - x[[1]] - x[[2]] - x[[3]]) * (1 - x[[1]] - x[[2]] - x[[3]])};
```

And genotype fitness be described by

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```
In[•]:= w = {{0, α, 0}, {β, α + β - γ1, β - γ2}, {0, α - γ3, -γ4}};
        TableForm[w, TableHeadings → {{"bb", "bB", "BB"}, {"aa", "aA", "AA"}}]
```

Out[•]//TableForm=

	aa	aA	AA
bb	0	α	0
bB	β	$\alpha+\beta-\gamma 1$	$\beta-\gamma 2$
BB	0	$\alpha-\gamma 3$	$-\gamma 4$

```
In[•]:= wgeno = {{0, β, α, α + β - γ1}, {β, 0, α + β - γ1, α - γ3},
           {α, α + β - γ1, 0, β - γ2}, {α + β - γ1, α - γ3, β - γ2, -γ4}};
        TableForm[wgeno,
         TableHeadings → {{"ab", "aB", "Ab", "AB"}, {"ab", "aB", "Ab", "AB"}}]
```

Out[•]//TableForm=

	ab	aB	Ab	AB
ab	0	β	α	$\alpha+\beta-\gamma 1$
aB	β	0	$\alpha+\beta-\gamma 1$	$\alpha-\gamma 3$
Ab	α	$\alpha+\beta-\gamma 1$	0	$\beta-\gamma 2$
AB	$\alpha+\beta-\gamma 1$	$\alpha-\gamma 3$	$\beta-\gamma 2$	$-\gamma 4$

With marginal fitness values given by x·w as follows:

```
In[•]:= wmarginal = x.wgeno;
        TableForm[wmarginal, TableHeadings → {{"w1", "w2", "w3", "w4"}, {" "}}]
```

Out[•]//TableForm=

w1	$x3\,\alpha + x2\,\beta + x4\,(\alpha+\beta-\gamma 1)$
w2	$x1\,\beta + x3\,(\alpha+\beta-\gamma 1) + x4\,(\alpha-\gamma 3)$
w3	$x1\,\alpha + x2\,(\alpha+\beta-\gamma 1) + x4\,(\beta-\gamma 2)$
w4	$x1\,(\alpha+\beta-\gamma 1) + x3\,(\beta-\gamma 2) + x2\,(\alpha-\gamma 3) - x4\,\gamma 4$

And mean fitness wbar:

```
In[•]:= wbar = Total[x * wmarginal] // Simplify
```

Out[•]=

$$2\,x1\,(x3\,\alpha + x2\,\beta + x4\,(\alpha+\beta-\gamma 1)) + 2\,x2\,(x3\,(\alpha+\beta-\gamma 1) + x4\,(\alpha-\gamma 3)) - x4\,(-2\,x3\,\beta + 2\,x3\,\gamma 2 + x4\,\gamma 4)$$

```
In[•]:= wbar // TeXForm
```

Then, the dynamics of the system can be described by :

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In[•]:=
```
dxdt = x;
dxdt[[1]] = x[[1]] (wmarginal[[1]] - wbar - m2 - m3) - r * DAB;
dxdt[[2]] = x[[2]] (wmarginal[[2]] - wbar - m2 - m3) + r * DAB + m2;
dxdt[[3]] = x[[3]] (wmarginal[[3]] - wbar - m2 - m3) + r * DAB + m3;
dxdt[[4]] = x[[4]] (wmarginal[[4]] - wbar - m2 - m3) - r * DAB;
dxdt // Simplify
```

Out[•]=
$$\{r\,(x2\,x3 - x1\,x4) + x1\,(-m2 - m3 + x3\,\alpha + x2\,\beta - 2\,x1\,(x3\,\alpha + x2\,\beta + x4\,(\alpha + \beta - \gamma1)) + x4\,(\alpha + \beta - \gamma1) - 2\,x2\,(x3\,(\alpha + \beta - \gamma1) + x4\,(\alpha - \gamma3)) + x4\,(-2\,x3\,\beta + 2\,x3\,\gamma2 + x4\,\gamma4)),\ m2 + r\,(-x2\,x3 + x1\,x4) + x2\,(-m2 - m3 + x1\,\beta - 2\,x1\,(x3\,\alpha + x2\,\beta + x4\,(\alpha + \beta - \gamma1)) + x3\,(\alpha + \beta - \gamma1) - 2\,x2\,(x3\,(\alpha + \beta - \gamma1) + x4\,(\alpha - \gamma3)) + x4\,(\alpha - \gamma3) + x4\,(-2\,x3\,\beta + 2\,x3\,\gamma2 + x4\,\gamma4)),\ m3 + r\,(-x2\,x3 + x1\,x4) + x3\,(-m2 - m3 + x1\,\alpha - 2\,x1\,(x3\,\alpha + x2\,\beta + x4\,(\alpha + \beta - \gamma1)) + x2\,(\alpha + \beta - \gamma1) + x4\,(\beta - \gamma2) - 2\,x2\,(x3\,(\alpha + \beta - \gamma1) + x4\,(\alpha - \gamma3)) + x4\,(-2\,x3\,\beta + 2\,x3\,\gamma2 + x4\,\gamma4)),\ r\,(x2\,x3 - x1\,x4) + x4\,(-m2 - m3 - 2\,x1\,(x3\,\alpha + x2\,\beta + x4\,(\alpha + \beta - \gamma1)) + x1\,(\alpha + \beta - \gamma1) + x3\,(\beta - \gamma2) - 2\,x2\,(x3\,(\alpha + \beta - \gamma1) + x4\,(\alpha - \gamma3)) + x2\,(\alpha - \gamma3) - x4\,\gamma4 + x4\,(-2\,x3\,\beta + 2\,x3\,\gamma2 + x4\,\gamma4))\}$$

In[•]:=
```
(*Tight linkage*)
newdxdttl = dxdt /.
   {r → 0, m2 → 0, m3 → 0, β → α, γ1 → γ / 2, γ2 → γ, γ3 → γ, γ4 → 2 γ} // Simplify
```

Out[•]=
$$\left\{x1\left(2 + 4\,x1^2 - 3\,x2 + 2\,x2^2 - 3\,x3 + 2\,x3^2 + x1\,(-6 + 4\,x2 + 4\,x3)\right)\alpha + \frac{1}{2}\,x1\left(3 + 2\,x1^2 - 3\,x3 + x2\,(-3 + 2\,x3) + x1\,(-5 + 2\,x2 + 2\,x3)\right)\gamma,\ \frac{1}{2}\,x2\left(\left(2 + 8\,x1^2 - 6\,x2 + 4\,x2^2 - 2\,x3 + 4\,x3^2 + 8\,x1\,(-1 + x2 + x3)\right)\alpha + \left(2 + 2\,x1^2 + 2\,x2\,(-1 + x3) - 3\,x3 + 2\,x1\,(-2 + x2 + x3)\right)\gamma\right),\ \frac{1}{2}\,x3\left(\left(2 + 8\,x1^2 - 2\,x2 + 4\,x2^2 - 6\,x3 + 4\,x3^2 + 8\,x1\,(-1 + x2 + x3)\right)\alpha + \left(2 + 2\,x1^2 - 3\,x2 - 2\,x3 + 2\,x2\,x3 + 2\,x1\,(-2 + x2 + x3)\right)\gamma\right)\right\}$$

Where **r** is the recombination rate, **DAB** is the linkage disequilibrium, **m** is the migration rate, which is the sum of **m2** (migration from parental haplotype **aB**) and **m3** (migration from parental haplotype **Ab**).

And allele frequencies and LD dynamics can be expressed as :
{dA/dt, dB/dt, dD/dt}

In[•]:=
```
dynp = {dxdt[[3]] + dxdt[[4]],
        dxdt[[2]] + dxdt[[4]],
        dxdt[[1]] * x[[4]] + dxdt[[4]] * x[[1]] - dxdt[[2]] * x[[3]] - dxdt[[3]] * x[[2]]} /.
      x4 → 1 - x1 - x2 - x3 /. transferpRule // FullSimplify;
dynp
```

Out[•]=
$$\{m3 - m2\,pA - m3\,pA - ld\,((-1+2\,pB)\,\beta + \gamma1 + 2\,ld\,\gamma1) + ld\,(pB\,(2\,\gamma1 - \gamma3) + ld\,(2\,\gamma2 + \gamma3 - \gamma4)) + pA\,(\alpha + pB\,(2\,(-1+pB)\,\gamma1 - pB\,\gamma3) + ld^2\,(4\,\gamma1 - 2\,(\gamma2+\gamma3) + \gamma4) - 2\,ld\,((-2+4\,pB)\,\gamma1 + \gamma2 + pB\,(-2\,(\gamma2+\gamma3) + \gamma4))) - pA^2\,(3\,\alpha + pB\,(6\,(-1+pB)\,\gamma1 + 2\,\gamma2 + pB\,(-2\,\gamma2 - 3\,\gamma3 + \gamma4)) - 2\,ld\,((-2+4\,pB)\,\gamma1 + \gamma2 + pB\,(-2\,(\gamma2+\gamma3) + \gamma4))) + pA^3\,(2\,\alpha + pB\,(4\,(-1+pB)\,\gamma1 + 2\,\gamma2 + pB\,(-2\,(\gamma2+\gamma3) + \gamma4))),$$
$$m2 - m2\,pB - m3\,pB + pB\,\beta - 3\,pB^2\,\beta + 2\,pB^3\,\beta + 4\,ld\,pB\,\gamma1 + 4\,ld^2\,pB\,\gamma1 - 2\,pA\,pB\,\gamma1 - 8\,ld\,pA\,pB\,\gamma1 + 2\,pA^2\,pB\,\gamma1 - 4\,ld\,pB^2\,\gamma1 + 6\,pA\,pB^2\,\gamma1 + 8\,ld\,pA\,pB^2\,\gamma1 - 6\,pA^2\,pB^2\,\gamma1 - 4\,pA\,pB^3\,\gamma1 + 4\,pA^2\,pB^3\,\gamma1 + ld\,(\alpha - 2\,pA\,\alpha + (-1-2\,ld+2\,pA)\,\gamma1) + ld^2\,\gamma2 - ld\,pA\,\gamma2 - 2\,ld^2\,pB\,\gamma2 + 4\,ld\,pA\,pB\,\gamma2 - pA^2\,pB\,\gamma2 - 4\,ld\,pA\,pB^2\,\gamma2 + 3\,pA^2\,pB^2\,\gamma2 - 2\,pA^2\,pB^3\,\gamma2 + 2\,ld^2\,\gamma3 - 2\,ld\,pB\,\gamma3 - 2\,ld^2\,pB\,\gamma3 + 4\,ld\,pA\,pB\,\gamma3 + 2\,ld\,pB^2\,\gamma3 - 2\,pA\,pB^2\,\gamma3 - 4\,ld\,pA\,pB^2\,\gamma3 + 2\,pA^2\,pB^2\,\gamma3 + 2\,pA\,pB^3\,\gamma3 - 2\,pA^2\,pB^3\,\gamma3 + (-1+pB)\,(ld + pA\,pB)^2\,\gamma4,$$
$$m2\,pA\,(-1+pB) + ld^3\,(4\,\gamma1 - 2\,(\gamma2+\gamma3) + \gamma4) + ld^2\,((-1+2\,pA)\,(-1+2\,pB)\,\gamma1 + \gamma2 + \gamma3 - \gamma4 + pB\,(-2\,\gamma2 + \gamma3 + \gamma4) + pA\,(\gamma2 - 2\,pB\,\gamma2 + (1+pB)\,(-2\,\gamma3 + \gamma4))) - ld\,(m2 + m3 + r - (1-2\,pA)^2\,\alpha - (1-2\,pB)^2\,\beta + \gamma1 + (-1+pB)\,pB\,(2\,\gamma1 - \gamma3) + pA^2\,((2 + 4\,(-1+pB)\,pB)\,\gamma1 + (-1 - 2\,(-1+pB)\,pB)\,\gamma2 + (-2+pB)\,pB\,(-2\,\gamma3 + \gamma4)) + pA\,(-2\,\gamma1 - 4\,(-1+pB)\,pB\,\gamma1 + \gamma2 + 2\,pB\,(2\,(-1+pB)\,\gamma2 + (-2+pB)\,\gamma3 + \gamma4 - pB\,\gamma4))) + (-1+pA)\,pB\,(m3 - pA\,(-1+pB)\,((-1+2\,pA)\,(-1+2\,pB)\,\gamma1 + pB\,\gamma3 + pA\,(\gamma2 - 2\,pB\,\gamma2 - 2\,pB\,\gamma3 + pB\,\gamma4)))\}$$

In[•]:=
```
(*Reducing dxdt*)
dxdt = dxdt[[1 ;; 3]] /. x4 → 1 - x1 - x2 - x3 // Simplify;
dxdt
```

Out[•]=
$$\{r\,(x1^2 + x2\,x3 + x1\,(-1+x2+x3)) + x1\,(-m2 - m3 + x3\,\alpha + x2\,\beta - 2\,x1\,(x3\,\alpha + x2\,\beta - (-1+x1+x2+x3)\,(\alpha+\beta-\gamma1)) - 2\,x2\,x3\,(\alpha+\beta-\gamma1) - (-1+x1+x2+x3)\,(\alpha+\beta-\gamma1) + 2\,x2\,(-1+x1+x2+x3)\,(\alpha-\gamma3) + (-1+x1+x2+x3)\,((-1+x1+x2)\,\gamma4 + x3\,(2\,\beta - 2\,\gamma2 + \gamma4))),$$
$$m2 - r\,(x1^2 + x2\,x3 + x1\,(-1+x2+x3)) + x2\,(-m2 - m3 + x1\,\beta - 2\,x1\,(x3\,\alpha + x2\,\beta - (-1+x1+x2+x3)\,(\alpha+\beta-\gamma1)) + x3\,(\alpha+\beta-\gamma1) - 2\,x2\,x3\,(\alpha+\beta-\gamma1) - (-1+x1+x2+x3)\,(\alpha-\gamma3) + 2\,x2\,(-1+x1+x2+x3)\,(\alpha-\gamma3) + (-1+x1+x2+x3)\,((-1+x1+x2)\,\gamma4 + x3\,(2\,\beta - 2\,\gamma2 + \gamma4))),$$
$$m3 - r\,(x1^2 + x2\,x3 + x1\,(-1+x2+x3)) + x3\,(-m2 - m3 + x1\,\alpha - 2\,x1\,(x3\,\alpha + x2\,\beta - (-1+x1+x2+x3)\,(\alpha+\beta-\gamma1)) + x2\,(\alpha+\beta-\gamma1) - 2\,x2\,x3\,(\alpha+\beta-\gamma1) - (-1+x1+x2+x3)\,(\beta-\gamma2) + 2\,x2\,(-1+x1+x2+x3)\,(\alpha-\gamma3) + (-1+x1+x2+x3)\,((-1+x1+x2)\,\gamma4 + x3\,(2\,\beta - 2\,\gamma2 + \gamma4)))\}$$

Finding solutions for the haplotype dynamics:
First, define a function for the dynamics

In[•]:=
```
xdots[{A_, B_, C_, R_, M_}][{X1_, X2_, X3_}] :=
  dxdt /. {α → A, β → B, r → R, x1 → X1, x2 → X2, x3 → X3} /.
    Thread[{γ1, γ2, γ3, γ4} → C ] /. Thread[{m2, m3} → M ];
pdots[{A_, B_, C_, R_, M_}][{PA_, PB_, LD_}] :=
  dynp /. {α → A, β → B, r → R, pA → PA, pB → PB, ld → LD} /.
    Thread[{γ1, γ2, γ3, γ4} → C] /. Thread[{m2, m3} → M ];

xdots1[{A_, B_, C_, R_, M_}][{X1_}] :=
  dxdt[[1]] /. {α → A, β → B, r → R, x1 → X1} /. Thread[{γ1, γ2, γ3, γ4} → C ] /.
   Thread[{m2, m3} → M ];
xdots2[{A_, B_, C_, R_, M_}][{X1_, X2_}] :=
  dxdt[[2]] /. {α → A, β → B, r → R, x1 → X1, x2 → X2} /.
    Thread[{γ1, γ2, γ3, γ4} → C ] /. Thread[{m2, m3} → M ];
xdots3[{A_, B_, C_, R_, M_}][{X1_, X2_, X3_}] :=
  dxdt[[3]] /. {α → A, β → B, r → R, x1 → X1, x2 → X2, x3 → X3} /.
    Thread[{γ1, γ2, γ3, γ4} → C ] /. Thread[{m2, m3} → M ];
```

In[•]:=
```
NSolve[
 xdots[{0.01, 0.01, {0.001, 0.002, 0.002, 0.004}, 0.2, {0, 0}}][{x1, x2, x3}] ==
  {0, 0, 0}, {x1, x2, x3}, Reals]
```

Out[•]=
```
{{x1 → 200.29283, x2 → -199.74283, x3 → -199.74283},
 {x1 → 6.6933174, x2 → -4.6279629, x3 → -4.6279629},
 {x1 → -4.4886527, x2 → 3.3920925, x3 → 6.6328383},
 {x1 → -4.4886527, x2 → 6.6328383, x3 → 3.3920925},
 {x1 → 3.2892105, x2 → -4.3071511, x3 → -4.3071511}, {x1 → 0., x2 → 0., x3 → 1.},
 {x1 → 0., x2 → 1., x3 → 0.}, {x1 → 1., x2 → 0., x3 → 0.},
 {x1 → 0., x2 → 0., x3 → 0.6}, {x1 → 0., x2 → 0.6, x3 → 0.},
 {x1 → 0.29720772, x2 → 0.24824169, x3 → 0.24824169}, {x1 → 0.5, x2 → 0., x3 → 0.5},
 {x1 → 0.5, x2 → 0.5, x3 → 0.}, {x1 → 0., x2 → 0., x3 → 0.}}
```

In[•]:=
```
pdotsLE[{A_,B_,C_,R_,M_}][{PA_,PB_,LD_}]:= dynp/.{α→A, β→B, r→R, pA→PA, pB→PB, ld→0}
SolspLE[{A_,B_,C_,R_,M_}]:=Solve[pdots[{A,B,C,R,M}][{pA,pB,0}][[1;;2]]=={0,0}, {pA,pB}];
SolspLEN[{A_,B_,C_,R_,M_}]:=NSolve[pdots[{A,B,C,R,M}][{pA,pB,0}][[1;;2]]=={0,0}, {pA,pB}]
```

Numerical equilibria

In[ ]:=
```
Solsx[{A_, B_, C_, R_, M_}] :=
 Solve[xdots[{A, B, C, R, M}][{x1, x2, x3}] == {0, 0, 0}, {x1, x2, x3}]
Solsp[{A_, B_, C_, R_, M_}] :=
 Solve[pdots[{A, B, C, R, M}][{pA, pB, ld}] == {0, 0, 0}, {pA, pB, ld}]

(*Solutions for each equation*)
Solsx1[{A_, B_, C_, R_, M_}] := Solve[xdots1[{A, B, C, R, M}][{x1}] == {0}, x1]
Solsx2[{A_, B_, C_, R_, M_}] := Solve[xdots2[{A, B, C, R, M}][{x2}] == {0}, {x2}]
Solsx3[{A_, B_, C_, R_, M_}] := Solve[xdots[{A, B, C, R, M}][{x3}] == {0}, {x3}]
```

Calculate the Jacobian matrix

In[ ]:=
```
Jacx[{A_, B_, C_, R_, M_}][x_] :=
 D[xdots[{A, B, C, R, M}][{x1, x2, x3}], {{x1, x2, x3}}] /. x
Jacp[{A_, B_, C_, R_, M_}][p_] :=
 D[pdots[{A, B, C, R, M}][{pA, pB, ld}], {{pA, pB, ld}}] /. p
Jacp[{A_, B_, C_, LE, M_}][p_] :=
 D[pdots[{A, B, C, _, M}][{pA, pB, 0}][[1 ;; 2]], {{pA, pB}}] /. p
```

Test plot of dynamics

Eigenvalues at monomorphic and polymorphic ((x1==x4) || (x2==x3) = 1/2) equilibria population (Codominant DMI)

Eigenvalues at monomorphic and polymorphic ((x1 == x4) || (x2 == x3) = 1/2) equilibria population (Recesive DMI)

Special cases by haplotype freq

$\alpha = \beta$; $\gamma = 0$; m = 0; r=0

α != β; $\gamma = 0$; m = 0; r=0

$\alpha = \beta$; $\gamma = 0$; m = 0; r!=0

α != β; $\gamma = 0$; m = 0; r!=0

$\alpha = \beta$; $\gamma = 0$; m =! 0; r=0

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Special cases by allele freq

LE

Recessive DMIs; $\alpha == \beta$; m=0; No polymorphic stable equilibria

Recessive DMIs; α != β ; m=0 (Higher order solutions)

Codominant DMIs; $\alpha == \beta$; m=0

Codominant DMIs; α != β; m=0 (1 polymorphic equilibrium)

Recessive DMIs; M > 0 (Higer order solutions)

LE; migration

No DMIs

Codominant DMIs ($\alpha == \beta$) (Higher order terms when m!=0)

Recessive DMIs ($\alpha == \beta$) (No solutions)

Phase plane plot: Recessive

Phase plane plot: Codominant

LDE

$\alpha == \beta$, No DMIs -> Stable for r=0 or α=r/2 (haplotype dynamics)

α != β, No DMIs -> Stable for pA=1/2; pB=1/2 (allele dynamics)

Codominant DMIs, no migration, and tight linkage (r=0) -> 1 polymorphic equilibrium

In[ ]:=
```
Γx = {γ / 2, γ, γ, 2 γ};
M = {0, 0};
(*sols = Solsp[{α,β,Γx,0,M}];
sols*)
```

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In[•]:=
```
solHap = Solsx[{α, β, Γx, 0, M}][[All, All, All]] // Simplify;
solHap
```

Out[•]=

$$\left\{\{x1 \to 0, x2 \to 0, x3 \to 1\}, \left\{x1 \to 0, x2 \to 0, x3 \to \frac{\beta+\gamma}{2\,\beta}\right\}, \left\{x1 \to 0, x2 \to \frac{1}{2}, x3 \to \frac{1}{2}\right\},\right.$$
$$\left\{x1 \to 0, x2 \to \frac{8\,\alpha\,\beta+6\,\beta\,\gamma-2\,\gamma^2}{16\,\alpha\,\beta-\gamma^2}, x3 \to \frac{8\,\alpha\,\beta+6\,\alpha\,\gamma-2\,\gamma^2}{16\,\alpha\,\beta-\gamma^2}\right\}, \{x1 \to 0, x2 \to 1, x3 \to 0\},$$
$$\left\{x1 \to 0, x2 \to \frac{\alpha+\gamma}{2\,\alpha}, x3 \to 0\right\}, \left\{x1 \to \frac{1}{2}, x2 \to 0, x3 \to \frac{1}{2}\right\}, \left\{x1 \to \frac{1}{2}, x2 \to \frac{1}{2}, x3 \to 0\right\},$$
$$\{x1 \to 1, x2 \to 0, x3 \to 0\}, \left\{x1 \to \frac{2\,\alpha+2\,\beta+3\,\gamma}{4\,\alpha+4\,\beta+2\,\gamma}, x2 \to 0, x3 \to 0\right\},$$
$$\left\{x1 \to \frac{(2\,\alpha+2\,\beta-\gamma)\,\gamma}{16\,\alpha\,\beta-\gamma^2}, x2 \to \frac{\alpha\,(8\,\beta-2\,\gamma)}{16\,\alpha\,\beta-\gamma^2}, x3 \to \frac{8\,\alpha\,\beta-2\,\beta\,\gamma}{16\,\alpha\,\beta-\gamma^2}\right\},$$
$$\left\{x1 \to \frac{8\,\alpha\,\beta+2\,\alpha\,\gamma-2\,\gamma^2}{16\,\alpha\,\beta-\gamma^2}, x2 \to -\frac{\gamma\,(-2\,\alpha+6\,\beta+\gamma)}{-16\,\alpha\,\beta+\gamma^2}, x3 \to 0\right\},$$
$$\left.\left\{x1 \to \frac{8\,\alpha\,\beta+2\,(\beta-\gamma)\,\gamma}{16\,\alpha\,\beta-\gamma^2}, x2 \to 0, x3 \to -\frac{\gamma\,(6\,\alpha-2\,\beta+\gamma)}{-16\,\alpha\,\beta+\gamma^2}\right\}, \{x1 \to 0, x2 \to 0, x3 \to 0\}\right\}$$

Test for stability

In[•]:=
```
For[i = 1, i ≤ Length[solHap], i++,
 isreal =
  Reduce[solHap[[i]][[All, 2]] ∈ Reals && γ ≥ 0 && α ≥ 0 && β ≥ 0, {α, γ, β}, Reals];
 If[isreal ≠ False, , ,
  ei = Eigenvalues[Jacx[{α, β, Γx, 0, M}][solHap[[i]]]] // Simplify;
  (*(Check if eigenvalue is stable, i.e. real and negative)*)
  isstable = Reduce[ei < {0, 0, 0} && α ≥ 0 && β ≥ 0 &&
      γ ≥ 0 && 1 ≥ solHap[[i]][[All, 2]] ≥ 0, {α, β}, Reals ] // Simplify;
  If[isstable == False,
   Print[solHap[[i]] // Simplify, "\t Unstable"],
   Print[0], Print[solHap[[i]], "\t Stable (i=", i, ") \n",
     Style[Reduce[1 ≥ solHap[[i]][[All, 2]] ≥ 0 && β ≥ 0 && α ≥ 0 &&
        γ ≥ 0 && ei < {0, 0, 0}, {α, γ}, Reals], Bold]];
  ];
 ];
]
```

$\{x1 \to 0, x2 \to 0, x3 \to 1\}$ Unstable

$\left\{x1 \to 0, x2 \to 0, x3 \to \frac{\beta+\gamma}{2\beta}\right\}$ Unstable

$\left\{x1 \to 0, x2 \to \frac{1}{2}, x3 \to \frac{1}{2}\right\}$ Unstable

$\left\{x1 \to 0, x2 \to \frac{8\alpha\beta+6\beta\gamma-2\gamma^2}{16\alpha\beta-\gamma^2}, x3 \to \frac{8\alpha\beta+6\alpha\gamma-2\gamma^2}{16\alpha\beta-\gamma^2}\right\}$ Unstable

$\{x1 \to 0, x2 \to 1, x3 \to 0\}$ Unstable

$\left\{x1 \to 0, x2 \to \frac{\alpha+\gamma}{2\alpha}, x3 \to 0\right\}$ Unstable

$\left\{x1 \to \frac{1}{2}, x2 \to 0, x3 \to \frac{1}{2}\right\}$ Stable (i=7)

$\beta \geq 0$ && $\alpha > 0$ && $\gamma > 4\beta$

$\left\{x1 \to \frac{1}{2}, x2 \to \frac{1}{2}, x3 \to 0\right\}$ Stable (i=8)

$\beta > 0$ && $\alpha \geq 0$ && $\gamma > 4\alpha$

$\{x1 \to 1, x2 \to 0, x3 \to 0\}$ Unstable

$\left\{x1 \to \frac{2\alpha+2\beta+3\gamma}{4\alpha+4\beta+2\gamma}, x2 \to 0, x3 \to 0\right\}$ Unstable

$\left\{x1 \to \frac{(2\alpha+2\beta-\gamma)\gamma}{16\alpha\beta-\gamma^2}, x2 \to \frac{\alpha(8\beta-2\gamma)}{16\alpha\beta-\gamma^2}, x3 \to \frac{8\alpha\beta-2\beta\gamma}{16\alpha\beta-\gamma^2}\right\}$ Stable (i=11)

$\beta > 0$ && (($0 < \alpha \leq \beta$ && $0 < \gamma < 4\alpha$) || ($\alpha > \beta$ && $0 < \gamma < 4\beta$))

$\left\{x1 \to \frac{8\alpha\beta+2\alpha\gamma-2\gamma^2}{16\alpha\beta-\gamma^2}, x2 \to -\frac{\gamma(-2\alpha+6\beta+\gamma)}{-16\alpha\beta+\gamma^2}, x3 \to 0\right\}$ Unstable

$\left\{x1 \to \frac{8\alpha\beta+2(\beta-\gamma)\gamma}{16\alpha\beta-\gamma^2}, x2 \to 0, x3 \to -\frac{\gamma(6\alpha-2\beta+\gamma)}{-16\alpha\beta+\gamma^2}\right\}$ Unstable

$\{x1 \to 0, x2 \to 0, x3 \to 0\}$ Unstable

Showing the solution: $\{x1 \to \frac{(2\alpha+2\beta-\gamma)\gamma}{16\alpha\beta-\gamma^2}, x2 \to \frac{\alpha(8\beta-2\gamma)}{16\alpha\beta-\gamma^2}, x3 \to \frac{8\alpha\beta-2\beta\gamma}{16\alpha\beta-\gamma^2}\}$

In[•]:=
```
conditions = Reduce[1 ≥ solHap⟦11⟧⟦All, 2⟧ ≥ 0 && β ≥ 0 && α ≥ 0 && γ ≥ 0 &&
    Eigenvalues[Jacx[{α, β, rx, 0, M}][solHap⟦11⟧]] < {0, 0, 0}, {α, γ}, Reals];
TraditionalForm[
 conditions //. Or → Composition[(Column[#, Right, Frame → All] &), List]]
```

Out[•]//TraditionalForm=

$$\beta > 0 \wedge \begin{array}{|c|}\hline 0 < \alpha \leq \beta \wedge 0 < \gamma < 4\alpha \\ \hline \alpha > \beta \wedge 0 < \gamma < 4\beta \\ \hline \end{array}$$

In[•]:=
```
haplotoallele[solHap⟦11⟧⟦All, 2⟧]
```

Out[•]=

$$\left\{\frac{8\alpha\beta-2\beta\gamma}{16\alpha\beta-\gamma^2}, \frac{\alpha(8\beta-2\gamma)}{16\alpha\beta-\gamma^2}, -\frac{4\alpha\beta(4\alpha-\gamma)(4\beta-\gamma)}{\left(-16\alpha\beta+\gamma^2\right)^2}\right\}$$

```
In[•]:= (*In terms of allele frequencies*)
        haplotoallele[solHap[[11]][[All, 2]]] /. {β → α} // Simplify
```

Out[•]=

$$\left\{\frac{2\,\alpha}{4\,\alpha+\gamma},\ \frac{2\,\alpha}{4\,\alpha+\gamma},\ -\frac{4\,\alpha^2}{(4\,\alpha+\gamma)^2}\right\}$$

Recessive DMIs, no migration, and tight linkage (r=0) -> All equilibria unstable

---

Perturbation analysis

Weak migration (m -> 0) (Codominant; LE)

Weak migration (m -> 0) (Codominant; LDE; r=0)

Weak recombination (r -> 0) (Codominant; LDE; assuming α=β; m=0)

```
In[•]:= oldsol = {pA → (8 α β - 2 β γ)/(16 α β - γ^2), pB → (α (8 β - 2 γ))/(16 α β - γ^2), ld → -(4 α β (4 α - γ) (4 β - γ))/(-16 α β + γ^2)^2};
```

```
In[•]:= oldsol = oldsol /. β → α // Simplify
```

Out[•]=

$$\left\{\text{pA} \to \frac{2\,\alpha}{4\,\alpha+\gamma},\ \text{pB} \to \frac{2\,\alpha}{4\,\alpha+\gamma},\ \text{ld} \to -\frac{4\,\alpha^2}{(4\,\alpha+\gamma)^2}\right\}$$

```
In[•]:= (*First, simplify by setting m2=m3=0 & α=β*)
        dynpsimplified = dynp /.
            {β → α, m2 → 0, m3 → 0, γ1 → γ / 2, γ2 → γ, γ3 → γ, γ4 → 2 γ, pA → pB} // Simplify;
```

And set r=0*

```
In[•]:= dynpsimplifiedr = dynpsimplified /. {r → 0} // Simplify;
```

```
In[•]:= (*Add the recombination term to the solution *)
        pBeq = 2 α/(4 α + γ) + r C1;

        LDeq = -4 α^2/(4 α + γ)^2 + r C2;
```

```
In[•]:= (*And we replace pB and ld in the original *)
        newdynp = dynpsimplified[[2 ;; 3]] /. {pB → pBeq, ld → LDeq} // Simplify;
```

```
In[•]:= (*Then we calculate the approximation as r-->0*)
        seriesr = Series[newdynp, {r, 0, 1}] // Normal;
```

In[•]:=
```
(*And solve for C and replace in the solution*)
newSolC = Solve[seriesr == {0, 0}, {C1, C2}] // Simplify
```

Out[•]=

$$\left\{\left\{C_1 \to -\frac{4\,\alpha\,(2\,\alpha+\gamma)}{\gamma\,(4\,\alpha+\gamma)^2},\ C_2 \to \frac{8\,\alpha^2\,(8\,\alpha+3\,\gamma)}{\gamma\,(4\,\alpha+\gamma)^3}\right\}\right\}$$

In[•]:=
```
newpB = pBeq /. newSolC[[1]]
newLD = LDeq /. newSolC[[1]]
```

Out[•]=

$$-\frac{4\,r\,\alpha\,(2\,\alpha+\gamma)}{\gamma\,(4\,\alpha+\gamma)^2}+\frac{2\,\alpha}{4\,\alpha+\gamma}$$

Out[•]=

$$-\frac{4\,\alpha^2}{(4\,\alpha+\gamma)^2}+\frac{8\,r\,\alpha^2\,(8\,\alpha+3\,\gamma)}{\gamma\,(4\,\alpha+\gamma)^3}$$

In[•]:=
```
solutionapprox = {pA → newpB, pB → newpB, ld → newLD}
```

Out[•]=

$$\left\{\mathrm{pA} \to -\frac{4\,r\,\alpha\,(2\,\alpha+\gamma)}{\gamma\,(4\,\alpha+\gamma)^2}+\frac{2\,\alpha}{4\,\alpha+\gamma},\ \mathrm{pB} \to -\frac{4\,r\,\alpha\,(2\,\alpha+\gamma)}{\gamma\,(4\,\alpha+\gamma)^2}+\frac{2\,\alpha}{4\,\alpha+\gamma},\ \mathrm{ld} \to -\frac{4\,\alpha^2}{(4\,\alpha+\gamma)^2}+\frac{8\,r\,\alpha^2\,(8\,\alpha+3\,\gamma)}{\gamma\,(4\,\alpha+\gamma)^3}\right\}$$

In[ ]:=
```
sol = {pA → -4 r α (2 α + γ)/(γ (4 α + γ)^2) + 2 α/(4 α + γ), ld → -4 α^2/(4 α + γ)^2 + 8 r α^2 (8 α + 3 γ)/(γ (4 α + γ)^3)};
Manipulate[Plot[{-4 r α (2 α + γ)/(γ (4 α + γ)^2) + 2 α/(4 α + γ), -4 α^2/(4 α + γ)^2 + 8 r α^2 (8 α + 3 γ)/(γ (4 α + γ)^3)},
  {γ, 0.001, 0.05}, PlotLegends → {"Freq", "LD"}, AxesLabel → Automatic],
 {r, 0.00001, 0.001}, {α, 0.001, 0.1}]
```

Out[ ]=




```
conditions = Reduce[1 ≥ newpB ≥ 0 && α ≥ 0 && γ ≥ 0 && 1 / 2 ≥ r ≥ 0, {α, γ, r}, Reals];
TraditionalForm[
 conditions //. Or → Composition[(Column[#, Right, Frame → All] &), List]]
```

Out[ ]//TraditionalForm=

	$\alpha = 0 \land \gamma > 0 \land 0 \le r \le \frac{1}{2}$
$\alpha > 0 \land$	$0 < \gamma < \frac{1}{2}\sqrt{16\alpha^2+1} + \frac{1}{2}(1-4\alpha) \land 0 \le r \le \frac{4\alpha\gamma+\gamma^2}{4\alpha+2\gamma}$
	$\gamma \ge \frac{1}{2}\sqrt{16\alpha^2+1} + \frac{1}{2}(1-4\alpha) \land 0 \le r \le \frac{1}{2}$

```
In[ ]:= conds = Reduce[newLD < 0 && α ≥ 0 && γ ≥ 0 && 1 / 2 ≥ r ≥ 0, {α, γ, r}, Reals];
TraditionalForm[
 conds //. Or → Composition[(Column[#, Right, Frame → All] &), List]]
```

Out[ ]//TraditionalForm=

$$\alpha > 0 \land \begin{array}{|r|}\hline 0 < \gamma \leq \frac{1}{2}\sqrt{16\alpha^2+8\alpha+9} + \frac{1}{2}(3-4\alpha) \land 0 \leq r < \frac{4\alpha\gamma+\gamma^2}{16\alpha+6\gamma} \\ \hline \gamma > \frac{1}{2}\sqrt{16\alpha^2+8\alpha+9} + \frac{1}{2}(3-4\alpha) \land 0 \leq r \leq \frac{1}{2} \\ \hline \end{array}$$

```
1/2 Sqrt[16 α^2 + 8 α + 9] + 1/2 (3 - 4 α) /. α → 0.001
```

Out[ ]=

```
2.9986679
```

```
In[ ]:= (4 α γ + γ^2)/(16 α + 6 γ) /. {α → 0.01, γ → 0.02}
```

Out[ ]=

```
0.0042857143
```

```
In[ ]:= solutionapprox /. {α → 0.002, γ → 0.004, r → {0, 0.00001}} // TableForm
```

Out[ ]//TableForm=

```
pA → {0.33333333, 0.33222222}
pB → {0.33333333, 0.33222222}
ld → {-0.11111111, -0.10981481}
```

Conditions for existence (But not stability)

```
In[ ]:= conditions = Reduce[1 ≥ newpB ≥ 0 && α ≥ 0 && γ ≥ 0 && 1 / 2 ≥ r ≥ 0, {α, γ, r}, Reals];
TraditionalForm[
 conditions //. Or → Composition[(Column[#, Right, Frame → All] &), List]]
```

Out[ ]//TraditionalForm=

$$\begin{array}{|l|}\hline \alpha = 0 \land \gamma > 0 \land 0 \leq r \leq \frac{1}{2} \\ \hline \alpha > 0 \land \begin{array}{|r|}\hline 0 < \gamma < \frac{1}{2}\sqrt{16\alpha^2+1} + \frac{1}{2}(1-4\alpha) \land 0 \leq r \leq \frac{4\alpha\gamma+\gamma^2}{4\alpha+2\gamma} \\ \hline \gamma \geq \frac{1}{2}\sqrt{16\alpha^2+1} + \frac{1}{2}(1-4\alpha) \land 0 \leq r \leq \frac{1}{2} \\ \hline \end{array} \\ \hline \end{array}$$

In[•]:=
```
(*And get the eigenvalues*)
eim = Eigenvalues[Jacp[{α, α, {γ / 2, γ, γ, 2 γ}, r, {0, 0}}][
    {pA → newpB, pB → newpB, ld → newLD}]] // Simplify
```

Out[•]=

$$\left\{\frac{\alpha\left(8\,r\,\alpha\,\gamma\,(4\,\alpha+\gamma)^3-(4\,\alpha-\gamma)\,\gamma^2\,(4\,\alpha+\gamma)^3+16\,r^2\,\alpha\,(2\,\alpha+\gamma)^2\,(6\,\alpha+\gamma)\right)}{\gamma^2\,(4\,\alpha+\gamma)^4},\right.$$

$$-\frac{1}{4\,\gamma^2\,(4\,\alpha+\gamma)^4}\left(384\,\alpha^4\,\gamma^3+352\,\alpha^3\,\gamma^4+120\,\alpha^2\,\gamma^5+18\,\alpha\,\gamma^6+\right.$$
$$\gamma^7-32\,r^2\,\alpha^2\,(2\,\alpha+\gamma)^2\,(14\,\alpha+5\,\gamma)+2\,r\,\gamma\,(4\,\alpha+\gamma)^2\left(32\,\alpha^3+\gamma^3\right)+$$
$$\sqrt{\left(\gamma^6\,(2\,\alpha+\gamma)^2\,(4\,\alpha+\gamma)^6+1024\,r^4\,\alpha^4\,(2\,\alpha+\gamma)^4\left(4\,\alpha^2+12\,\alpha\,\gamma+\gamma^2\right)+\right.}$$
$$4\,r\,\gamma^4\,(4\,\alpha+\gamma)^5\left(-64\,\alpha^4-96\,\alpha^3\,\gamma-32\,\alpha^2\,\gamma^2+2\,\alpha\,\gamma^3+\gamma^4\right)-128\,r^3\,\alpha^2\,\gamma\,(2\,\alpha+\gamma)^2$$
$$\left(15360\,\alpha^6+14336\,\alpha^5\,\gamma+4416\,\alpha^4\,\gamma^2+352\,\alpha^3\,\gamma^3-64\,\alpha^2\,\gamma^4-14\,\alpha\,\gamma^5-\gamma^6\right)+$$
$$4\,r^2\,\gamma^2\,(4\,\alpha+\gamma)^2\left(49152\,\alpha^8+78848\,\alpha^7\,\gamma+52992\,\alpha^6\,\gamma^2+22016\,\alpha^5\,\gamma^3+\right.$$
$$\left.\left.\left.7552\,\alpha^4\,\gamma^4+2176\,\alpha^3\,\gamma^5+416\,\alpha^2\,\gamma^6+40\,\alpha\,\gamma^7+\gamma^8\right)\right)\right),$$

$$-\frac{1}{4\,\gamma^2\,(4\,\alpha+\gamma)^4}\left(384\,\alpha^4\,\gamma^3+352\,\alpha^3\,\gamma^4+120\,\alpha^2\,\gamma^5+18\,\alpha\,\gamma^6+\gamma^7-\right.$$
$$32\,r^2\,\alpha^2\,(2\,\alpha+\gamma)^2\,(14\,\alpha+5\,\gamma)+2\,r\,\gamma\,(4\,\alpha+\gamma)^2\left(32\,\alpha^3+\gamma^3\right)-$$
$$\sqrt{\left(\gamma^6\,(2\,\alpha+\gamma)^2\,(4\,\alpha+\gamma)^6+1024\,r^4\,\alpha^4\,(2\,\alpha+\gamma)^4\left(4\,\alpha^2+12\,\alpha\,\gamma+\gamma^2\right)+\right.}$$
$$4\,r\,\gamma^4\,(4\,\alpha+\gamma)^5\left(-64\,\alpha^4-96\,\alpha^3\,\gamma-32\,\alpha^2\,\gamma^2+2\,\alpha\,\gamma^3+\gamma^4\right)-128\,r^3\,\alpha^2\,\gamma\,(2\,\alpha+\gamma)^2$$
$$\left(15360\,\alpha^6+14336\,\alpha^5\,\gamma+4416\,\alpha^4\,\gamma^2+352\,\alpha^3\,\gamma^3-64\,\alpha^2\,\gamma^4-14\,\alpha\,\gamma^5-\gamma^6\right)+$$
$$4\,r^2\,\gamma^2\,(4\,\alpha+\gamma)^2\left(49152\,\alpha^8+78848\,\alpha^7\,\gamma+52992\,\alpha^6\,\gamma^2+22016\,\alpha^5\,\gamma^3+\right.$$
$$\left.\left.\left.\left.7552\,\alpha^4\,\gamma^4+2176\,\alpha^3\,\gamma^5+416\,\alpha^2\,\gamma^6+40\,\alpha\,\gamma^7+\gamma^8\right)\right)\right)\Bigg\}$$

In[•]:=
```
testvals = {r → 0.0001, α → 0.01, γ → 0.005} ;
oldsol /. testvals
{pA → newpB, pB → newpB, LD → newLD} /. testvals
```

Out[•]=
```
{pA → 0.44444444, pB → 0.44444444, ld → -0.19753086}
```

Out[•]=
```
{pA → 0.4345679, pB → 0.4345679, LD → -0.18085048}
```

In[•]:=
```
eim /. testvals // FullSimplify
ei = Eigenvalues[Jacp[{α, α, {γ / 2, γ, γ, 2 γ}, r, {0, 0}}][oldsol]] /. testvals
```

Out[•]=
```
{-0.0074158817, -0.0024804212, -0.0014318788}
```

Out[•]=
```
{-0.0077777778, -0.002511864, -0.0011992471}
```

Equilibrium is stable for this example

Numerical solutions for intermediate cases

QLE

```
In[ ]:= oldsol = {pA → α/(2 α + γ), ld → 0};

In[ ]:= (*First, simplify by setting m2=m3=0 & α=β*)
        dynpsimplified = dynp /.
            {β → α, m2 → 0, m3 → 0, γ1 → γ / 2, γ2 → γ, γ3 → γ, γ4 → 2 γ, pA → pB} // Simplify;

In[ ]:= (*Then, add the recombination term to the solution *)
        pBeq = α/(2 α + γ) + ϵ C1;
        LDeq = 0 + ϵ C2;

In[ ]:= newdynp = dynpsimplified[[2 ;; 3]] /. {pB → pBeq, ld → LDeq};

In[ ]:= seriesr = Series[newdynp, {ϵ, 0, 1}] // Normal;

In[ ]:= (*And solve for C and replace in the solution*)
        newSolC = Solve[seriesr == {0, 0}, {C1, C2}] // Simplify
```

Out[ ]=

$$\left\{\left\{C_1 \to \frac{\alpha \gamma^3 (\alpha+\gamma)}{4 (2\alpha+\gamma)\left(\alpha\gamma^3 + r (2\alpha+\gamma)^3\right)\epsilon}, C_2 \to -\frac{\alpha^2 \gamma (\alpha+\gamma)^2}{2 (2\alpha+\gamma)\left(\alpha\gamma^3 + r (2\alpha+\gamma)^3\right)\epsilon}\right\}\right\}$$

```
In[ ]:= newpB = pBeq /. newSolC /. ϵ → (1 / r)
        newLD = LDeq /. newSolC /. ϵ → (1 / r)
```

Out[ ]=

$$\left\{\frac{\alpha}{2\alpha+\gamma} + \frac{\alpha \gamma^3 (\alpha+\gamma)}{4 (2\alpha+\gamma)\left(\alpha\gamma^3 + r (2\alpha+\gamma)^3\right)}\right\}$$

Out[ ]=

$$\left\{-\frac{\alpha^2 \gamma (\alpha+\gamma)^2}{2 (2\alpha+\gamma)\left(\alpha\gamma^3 + r (2\alpha+\gamma)^3\right)}\right\}$$

Allele dynamics plots

Phase diagrams

Manuscript (Supplementary Fig. S7)

```
In[•]:= p = pdots[{0.01, 0.01, {0.018 / 2, 0.018, 0.018, 2 × 0.018}, LE, {0, 0}}][{pA, pB, 0}][[
          1 ;; 2]];
        ze = NSolve[p == {0, 0} && pA ≤ 1.0000001 && pB ≤ 1.0000001,
              {pA, pB}, NonNegativeReals][[All, All, 2]] // Quiet;

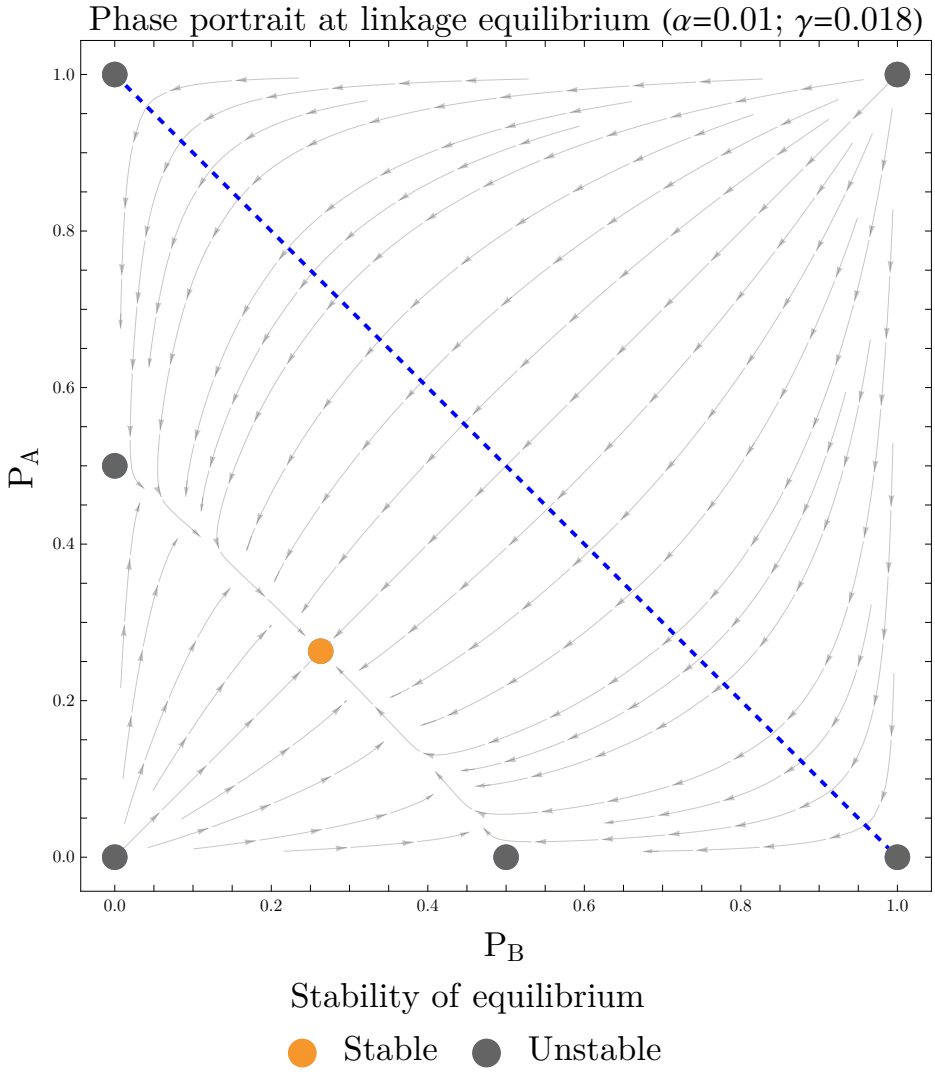
        eigenvs =
          Eigenvalues[Jacp[{0.01, 0.01, {0.018 / 2, 0.018, 0.018, 2 × 0.018}, LE, {0, 0}}][
              {pA → #[[1]], pB → #[[2]]}]] & /@ ze;

        stableeq = If[#[[1]] < 0 && #[[2]] < 0, 1, 0] & /@ eigenvs // Quiet;
        zeros = Join[ze[[#]], {stableeq[[#]]}] & /@ Range[Length[ze]] // Quiet;

        styleddata =
          Style[#, {RGBColor[245 / 255, 151 / 255, 45 / 255, 1], RGBColor[100 / 255, 100 / 255,
                100 / 255, 1]}[[Mod[#2, 2, 1]]]] &@@@ Transpose[{ze, stableeq}];


        Show[Legended[StreamPlot[p, {pA, 0, 1}, {pB, 0, 1},
            StreamStyle → {LightGray, Opacity[0.25]}, StreamColorFunction → "GrayTones",
            StreamColorFunctionScaling → False, StreamScale → 0.08,
            StreamPoints → 30, AspectRatio → Automatic, StreamMarkers → "Arrow",
            Frame → {{True, True}, {True, True}},
            FrameLabel → {{Style["P_A", 18, Black], None}, {Style["P_B", 18, Black], None}},
            FrameTicks → Automatic, AspectRatio → 1, ImageSize → {500, 500},
            PlotLabel → Style["Phase portrait at linkage equilibrium (α=0.01; γ=0.018)",
              18, Black], PlotRangePadding → Scaled[0.04]],
          Placed[PointLegend[{RGBColor[245 / 255, 151 / 255, 45 / 255, 1],
              RGBColor[100 / 255, 100 / 255, 100 / 255, 1]}, {Style["Stable", 18, Black],
              Style["Unstable", 18, Black]}, LegendMarkerSize → 30,
            LegendLayout → "Row", LegendLabel → Style["Stability of equilibrium",
              18, Black, FontFamily → "Latin Modern Math"]], Bottom] ],
         Plot[1 - x, {x, 0, 1}, PlotStyle → Directive[{Blue, Dashed}]],
         ListPlot[styleddata, BaseStyle → PointSize[0.03]]]
        Export["~/Downloads/phaseportrait_le.pdf", %]
```

Out[ ]=



Out[ ]=

```
~/Downloads/phaseportrait_le.pdf
```

```
In[•]:= p = pdots[{0.01, 0.01, {0.022 / 2, 0.022, 0.022, 2 × 0.022}, LE, {0, 0}}][{pA, pB, 0}][[
          1 ;; 2]];
        ze = NSolve[p == {0, 0} && pA ≤ 1.0000001 && pB ≤ 1.0000001,
              {pA, pB}, NonNegativeReals][[All, All, 2]] // Quiet;

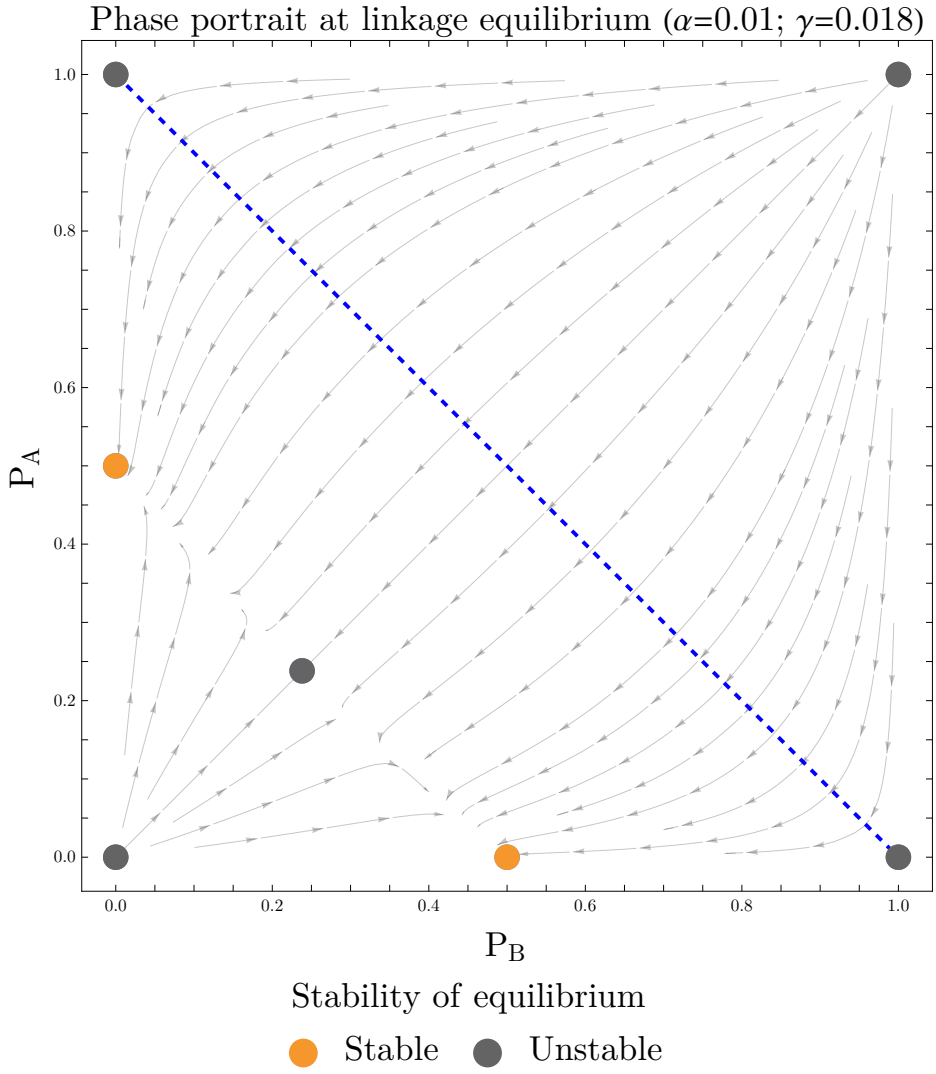
        eigenvs =
          Eigenvalues[Jacp[{0.01, 0.01, {0.022 / 2, 0.022, 0.022, 2 × 0.022}, LE, {0, 0}}][
              {pA → #[[1]], pB → #[[2]]}]] & /@ ze;

        stableeq = If[#[[1]] < 0 && #[[2]] < 0, 1, 0] & /@ eigenvs // Quiet;
        zeros = Join[ze[[#]], {stableeq[[#]]}] & /@ Range[Length[ze]] // Quiet;

        styleddata =
          Style[#, {RGBColor[245 / 255, 151 / 255, 45 / 255, 1], RGBColor[100 / 255, 100 / 255,
                100 / 255, 1]}[[Mod[#2, 2, 1]]]] &@@@ Transpose[{ze, stableeq}];

        Show[Legended[StreamPlot[p, {pA, 0, 1}, {pB, 0, 1},
            StreamStyle → {LightGray, Opacity[0.25]}, StreamColorFunction → "GrayTones",
            StreamColorFunctionScaling → False, StreamScale → 0.08,
            StreamPoints → 30, AspectRatio → Automatic, StreamMarkers → "Arrow",
            Frame → {{True, True}, {True, True}},
            FrameLabel → {{Style["P_A", 18, Black], None}, {Style["P_B", 18, Black], None}},
            FrameTicks → Automatic, AspectRatio → 1, ImageSize → {500, 500},
            PlotLabel → Style["Phase portrait at linkage equilibrium (α=0.01; γ=0.018)",
              18, Black], PlotRangePadding → Scaled[0.04]],
          Placed[PointLegend[{RGBColor[245 / 255, 151 / 255, 45 / 255, 1],
              RGBColor[100 / 255, 100 / 255, 100 / 255, 1]}, {Style["Stable", 18, Black],
              Style["Unstable", 18, Black]}, LegendMarkerSize → 30, LegendLayout → "Row",
             LegendLabel → Style["Stability of equilibrium", 18, Black]], Bottom] ],
         Plot[1 - x, {x, 0, 1}, PlotStyle → Directive[{Blue, Dashed}]],
         ListPlot[styleddata, BaseStyle → PointSize[0.03]]]
```

Out[ ]=



In[ ]:=
```
Export["~/Downloads/phaseportrait_le_2.pdf", %]
```

Out[ ]=

```
~/Downloads/phaseportrait_le_2.pdf
```

Full plots (All parameters, uncomment to run)

Codominant

```
In[•]:= (*Manipulate[
 p=pdots[{α,β,{γ/2,γ,γ,2γ},r,{m,m}}][{pA,pB,ld}];
 ze=NSolve[p=={0,0,0} && 0≤pA≤1.0000001&&0≤pB≤1.0000001 && 0.25≥ld≥-0.25,
     {pA,pB,ld}, Reals][[All,All,2]]//Quiet;
 eigenvs = Eigenvalues[
     Jacp[{α,β,{γ/2,γ,γ,2γ},r,{m,m}}][{pA→#[[1]], pB→#[[2]],ld →#[[3]]}]] & /@ ze;
 stableeq =If[#[[1]]<0&&#[[2]]<0&&#[[3]]<0,1,0]  & /@eigenvs//Quiet;
 zeros =Join[ze[[#]],{stableeq[[#]]}] &  /@ Range[Length[ze]]//Quiet;
 Show[ListPlot[zeros[[All,{1,2}]], AxesLabel→{"pA","pB"},
   PlotRange→{{-0.1,2.6},{-0.1,1.2}}, ImageSize→Large,
   PlotStyle→(If[#[[4]]==1,Red,Blue]&/@zeros),
   Epilog→{{Style[Text[zeros,{2.0,0.8}],10]}},
   PlotLabel→Style["Codominant DMIs",16, Black]],
  Graphics@Text[""<>ToString@#,#+{0,.05}]&/@zeros[[All,{1,2}]](*,
  StreamPlot[p,{pA,0,1}, {pB,0,1},
   StreamStyle→Opacity[0.5], PlotTheme→"Fine"]*)],
 {{α,0.01},0,1,0.01, Appearance→ "Labeled"},
  {{β,0.01},0,1,0.01,Appearance→ "Labeled"},
  {γ,0,1,0.001,Appearance→ "Labeled"},
 {m,0,1,0.001,Appearance→ "Labeled"},
 {r,0,0.5,0.01,Appearance→ "Labeled"},
 TrackedSymbols→{α,β,γ,m,r}]*)
```

Recessive DMIs

```
In[•]:= (*Manipulate[
 p=pdots[{α,β,{0,γ,γ,2γ},r,{m,m}}][{pA,pB,ld}];
 ze=NSolve[p=={0,0,0} && 0≤pA≤1.0000001&&0≤pB≤1.0000001 && 0.25≥ld≥-0.25,
     {pA,pB,ld}, Reals, WorkingPrecision→10][[All,All,2]]//Quiet;
 eigenvs = Eigenvalues[
     Jacp[{α,β,{0,γ,γ,2γ},r,{m,m}}][{pA→#[[1]], pB→#[[2]],ld →#[[3]]}]] & /@ ze;
 stableeq =If[#[[1]]<0&&#[[2]]<0&&#[[3]]<0,1,0]  & /@eigenvs//Quiet;
 zeros =Join[ze[[#]],{stableeq[[#]]}] &  /@ Range[Length[ze]]//Quiet;
 Show[ListPlot[zeros[[All,{1,2}]], AxesLabel→{"pA","pB"},
   PlotRange→{{-0.1,2.6},{-0.1,1.2}}, ImageSize→Large,
   PlotStyle→(If[#[[4]]==1,Red,Blue]&/@zeros),
   Epilog→{{Style[Text[zeros,{2.0,0.8}],10]}},
   PlotLabel→Style["Recessive DMIs",16, Black]],
  Graphics@Text[""<>ToString@#,#+{0,.05}]&/@zeros[[All,{1,2}]](*,
  StreamPlot[p,{pA,0,1}, {pB,0,1},
   StreamStyle→Opacity[0.5], PlotTheme→"Fine"]*)],
 {{α,0.01},0,1,0.01, Appearance→ "Labeled"},
  {{β,0.01},0,1,0.01,Appearance→ "Labeled"},
  {γ,0,1,0.001,Appearance→ "Labeled"},
 {m,0,1,0.01,Appearance→ "Labeled"},
 {r,0,0.5,0.01,Appearance→ "Labeled"},
 TrackedSymbols→{α,β,γ,m,r}]*)
```

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Evaluation of grid of recombination and DMI/OD

(Data for fig 2 A and 2 B in manuscript)

In[ ]:=
```
(*for reference:*)
TableForm[wgeno,
 TableHeadings → {{"ab", "aB", "Ab", "AB"}, {"ab", "aB", "Ab", "AB"}}]
```

Out[ ]//TableForm=

	ab	aB	Ab	AB
ab	0	β	α	$\alpha + \beta - \gamma 1$
aB	β	0	$\alpha + \beta - \gamma 1$	$\alpha - \gamma 3$
Ab	α	$\alpha + \beta - \gamma 1$	0	$\beta - \gamma 2$
AB	$\alpha + \beta - \gamma 1$	$\alpha - \gamma 3$	$\beta - \gamma 2$	$-\gamma 4$

In[ ]:=
```
A = 0.02;
B = A;
Γ = 0.01;
rec = 0;
migration = {0, 1 / 100000, 1 / 10000, 1 / 1000, 2 / 1000, 1 / 100} // N;
reclist = Join[{0}, 10 ^ Range[-5, -1, 0.5], {0.2, 0.3, 0.4, 0.5}] // N;
```

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In[ ]:=
```
(*Function to get all fully polymorphic stable equilibra*)
PolyEqs[{α_,β_,G_,r_,M_}] :=Module[{stablepoly, stableeqs,p,ze,eigenvs,stableeq},
    p=pdots[{α,β,G,r,M}][{pA,pB,ld}] ;
    ze=NSolve[p==={0,0,0} && 0≤pA≤1.0 && 0≤pB≤1.0 && Abs[ld]≤0.25,
    {pA,pB,ld},
    Reals,
    WorkingPrecision→30][[All,All,2]]//DeleteDuplicates // Quiet ;
    ze = DeleteDuplicates[ze, Abs[#1[[1]]-#2[[1]]]+Abs[#1[[2]]-#2[[2]]]+Abs[#1[[3]]-#2[[3]]]<0.000
    eigenvs = Eigenvalues[Jacp[{α,β,G,r,M}][{pA→#[[1]], pB→#[[2]], ld→#[[3]]}]] & /@ ze;
    stableeq = AllTrue[#,Negative]& /@ eigenvs;
    stableeqs=ze[[Position[stableeq,True]//Flatten]] ;
    stablepoly = Select[stableeqs,#[[1]]≠0 && #[[2]]≠0&];
    stablepoly = Select[stablepoly,And @@ NonNegative[alleletohaplo[#]]&] (*Get only p
]

PolyEqs[{α_,β_,G_,LE,M_}] :=Module[{stablepoly, stableeqs,p,ze,eigenvs,stableeq},
    p=pdots[{α,β,G,LE,M}][{pA,pB,0}] [[1;;2]];
    ze=NSolve[p==={0,0} && 0≤pA≤1.00000001&&0≤pB≤1.00000001,
    {pA,pB},
    Reals, WorkingPrecision→30][[All,All,2]]//DeleteDuplicates // Quiet ;
    ze = DeleteDuplicates[ze, Abs[#1[[1]]-#2[[1]]]+Abs[#1[[2]]-#2[[2]]]+Abs[#1[[3]]-#2[[3]]]<0.000
    eigenvs = Eigenvalues[Jacp[{α,β,G,LE,M}][{pA→#[[1]], pB→#[[2]]}]] & /@ ze;
    stableeq = AllTrue[#,Negative]& /@ eigenvs;
    stableeqs=ze[[Position[stableeq,True]//Flatten]] ;
    stablepoly = Select[stableeqs,#[[1]]≠0 && #[[2]]≠0&];
    stablepoly = Select[stablepoly,And @@ NonNegative[alleletohaplo[#]]&] (*Get only p
]

(*Function to get the nth element from the third sublist (i.e. values)*)
getElement[sublist_,element_] := Check[{sublist[[1]], sublist[[2]],sublist[[3,1,element]]},{

missingColourFunction[x_]:=If[MissingQ[x],None,ColorData["BlueGreenYellow"][x]];
```

Codominant heatmap data

LD

In[ ]:=
```
A = 0.01;
B = A;
Γ = 0.01;
rec = 0;
migration = {0};
(*migration = {0,0.00001, 0.0001,0.001,0.002, 0.01};*)
reclist = Join[{0}, 10 ^ Range[-7, -0.3, 0.125], {0.5}] // N;
```

```
In[•]:= allsolutionscodominant = {};
numsolcodominant = {};
sols1codominant = {};
For [m = 1, m ≤ Length[migration], m++,
 Print[migration⟦m⟧];
 For[Γ = 0.0005, Γ ≤ 5 A, Γ = Γ + 0.0005,
  For[i = 1, i ≤ Length[reclist], i++,
   stableeq =
    PolyEqs[{A, B, {Γ / 2, Γ, Γ, 2 Γ}, reclist⟦i⟧, {migration⟦m⟧, migration⟦m⟧}}];
   If[Length[stableeq] > 0, AppendTo[allsolutionscodominant,
      {Γ / A, reclist⟦i⟧, migration⟦m⟧, stableeq}];, False];
   If[Length[stableeq] == 1, AppendTo[sols1codominant, {Γ / A, reclist⟦i⟧,
       migration⟦m⟧, stableeq⟦1, 1⟧, stableeq⟦1, 3⟧}];, False];
   AppendTo[numsolcodominant,
    {Γ / A, reclist⟦i⟧, migration⟦m⟧, Length[stableeq]}];
  ]
 ]
]
0

In[•]:= SetDirectory[NotebookDirectory[]];
Export["../Data/codominant_sols_freqs_20250911_a=0.01.csv",
  sols1codominant, "CSV"];


(*Temporary for 5-10 A values*)
(*SetDirectory[NotebookDirectory[]];
Export["codominant_sols2.csv",numsolcodominant,"CSV"];*)

(*SetDirectory[NotebookDirectory[]];
Export["codominant_sols.csv",numsolcodominant,"CSV"];*)
```

With α=0.001

```
In[•]:= A = 0.001;
B = A;
Γ = 0.01;
rec = 0;
migration = {0};
(*migration = {0,0.00001, 0.0001,0.001,0.002, 0.01};*)
reclist = Join[{0}, 10 ^ Range[-7, -0.3, 0.125], {0.5}] // N;
```

```
In[•]:= allsolutionscodominant = {};
numsolcodominant = {};
sols1codominant = {};
For [m = 1, m ≤ Length[migration], m++,
 Print[migration[[m]];
 For[Γ = 0.00005, Γ ≤ 5 A, Γ = Γ + 0.00005,
  For[i = 1, i ≤ Length[reclist], i++,
   stableeq =
    PolyEqs[{A, B, {Γ / 2, Γ, Γ, 2 Γ}, reclist[[i]], {migration[[m]], migration[[m]]}}];
   If[Length[stableeq] > 0, AppendTo[allsolutionscodominant,
      {Γ / A, reclist[[i]], migration[[m]], stableeq}];, False];
   If[Length[stableeq] == 1, AppendTo[sols1codominant, {Γ / A, reclist[[i]],
       migration[[m]], stableeq[[1, 1]], stableeq[[1, 3]]}];, False];
   AppendTo[numsolcodominant,
    {Γ / A, reclist[[i]], migration[[m]], Length[stableeq]}];
  ]
 ]
]
0
```

```
In[•]:= SetDirectory[NotebookDirectory[]];
Export["../Data/codominant_sols_freqs_20250911_a=0.001.csv",
  sols1codominant, "CSV"];
```

Recessive

LD

```
In[•]:= A = 0.005;
B = A;
Γ = 0.01;
rec = 0;
migration = {0};
reclist = Join[10^Range[-6, -0.3, 0.12501], {0.5}] // N;
(*reclist = Join[10 ^ Range[-6,-0.3, 0.125],{0.5}] // N;*)

(*In general, 0.0025, but using 1/367 for a different list of gamma values*)
```

In[•]:=
```
allsolutionsrecessive = {};
numsolrecessive = {};
sols1recessive = {};
For [m = 1, m ≤ Length[migration], m++,
 Print[migration⟦m⟧];
 (*For[Γ = 0.0005, Γ≤16 A, Γ = Γ +0.00625,*)
 For[Γ = 0.0005, Γ ≤ 16 A, Γ = Γ + 0.001,
  nsolsgamma = 0;
  For[i = 1, i ≤ Length[reclist], i++,
   stableeq = PolyEqs[
     {A, B, {0, Γ, Γ, 2 Γ} // N, reclist⟦i⟧, {migration⟦m⟧, migration⟦m⟧}}];
   If[Length[stableeq] > 0,
    AppendTo[allsolutionsrecessive, {Γ / A, reclist⟦i⟧, migration⟦m⟧, stableeq}];
    If[Length[stableeq] > 1, Print[stableeq];
     Print[Γ / A], False], False];
   If[Length[stableeq] == 1, AppendTo[sols1recessive, {Γ / A, reclist⟦i⟧,
       migration⟦m⟧, stableeq⟦1, 1⟧, stableeq⟦1, 3⟧}];, False];
   AppendTo[numsolrecessive, {Γ / A, reclist⟦i⟧,
     migration⟦m⟧, Length[stableeq]}]; nsolsgamma = nsolsgamma + 1;
  ];
 ]
]
```

```
0
```

```
SetDirectory[NotebookDirectory[]];
Export["../Data/recessive_sols_ext_1.csv", sols1recessive, "CSV"];
```

In[ ]:=
```
A = 0.001;
B = A;
Γ = 0.001;
rec = 0;
migration = {0};
reclist = Join[10 ^ Range[-6, -0.3, 0.125], {0.5}] // N;
allsolutionsrecessive = {};
numsolrecessive = {};
sols1recessive = {};
For [m = 1, m ≤ Length[migration], m++,
 Print[migration[[m]]];
 For[Γ = 0.00005, Γ ≤ 16 A, Γ = Γ + 0.00025,
  For[i = 1, i ≤ Length[reclist], i++,
   stableeq =
    PolyEqs[{A, B, {0, Γ, Γ, 2 Γ}, reclist[[i]], {migration[[m]], migration[[m]]}}];
   If[Length[stableeq] > 0, AppendTo[allsolutionsrecessive,
      {Γ / A, reclist[[i]], migration[[m]], stableeq}];, False];
   If[Length[stableeq] == 1, AppendTo[sols1recessive, {Γ / A, reclist[[i]],
       migration[[m]], stableeq[[1, 1]], stableeq[[1, 3]]}];, False];
   AppendTo[numsolrecessive,
    {Γ / A, reclist[[i]], migration[[m]], Length[stableeq]}];
  ]
 ]
]


0

SetDirectory[NotebookDirectory[]];
Export["../Data/recessive_sols_ext_001.csv", sols1recessive, "CSV"];
```

Migration (Not used in Manuscript)

Dynamics in discrete time

Discrete time model (2 loci)